



Daniil Parshin

R&D Engineer | High-Performance C++ | Computer Vision

Profile

Performance-oriented R&D Engineer and Applied Mathematics student focused on high-performance C++ systems, computer vision, and model inference. Experienced in building CPU-side inference pipelines, profiling and optimization, ONNX Runtime-based solutions, and end-to-end tooling for training, evaluation, and deployment of detection models. Strong interest in low-level optimization, benchmarking, and production-oriented ML/CV engineering.

Employment Experience

Research Engineer

Lobachevsky Research Institute (LRI), Huawei — Nizhny Novgorod, Russia

June 2024 — Present

- Modernized the Mesa graphics driver by improving BC1 compression quality with dithering techniques and integrating BC7 support with ISPC-based acceleration and additional low-level optimizations.
- Built Python-based research tooling for automated trace processing, rendering-quality evaluation, and performance analysis.
- Developed a high-performance CPU inference library for ROI detection (text, faces, clothing, document regions) using ONNX Runtime, tiled large-frame inference, and OpenMP-based parallelization.
- Designed and maintained an end-to-end detection model pipeline for dataset conversion, dataset statistics, controlled preparation, Ultralytics YOLO training, per-class AP reporting, ONNX export/optimization, and benchmarking, producing deployable artifacts for the internal ROI inference library.

Resident and Ambassador, NEIMARK IT-Academy — Nizhny Novgorod, Russia

September 2022 — September 2024

[Program Page](#) | [Certificates](#)

- Represented NEIMARK as a resident and ambassador across educational and community initiatives.
- Delivered team case projects in algorithms and applied software development.
- Mentored Hackathon Academy participants; received an official letter of appreciation.
- Contributed to a winning School of Digital Ethics project and was recognized for impact in AI ethics initiatives.

Private tutor in Mathematics / Computer Science — Nizhny Novgorod, Russia

September 2022 — Present

- Prepared 30+ high school students for the Unified State Exam in specialized mathematics and computer science.
- Taught topics including algorithms, data structures, interpolation, nonlinear systems, and computational geometry.
- Developed structured learning plans and adapted explanations to different student backgrounds and skill levels.

Selected Projects

2022 — Present

- High-performance ROI detection library (C++, ONNX, OpenCV, OpenMP) | [GitHub](#)
- YOLO end-to-end training and deployment pipeline (Python, Shell) | [GitHub](#)
- Local-first audio/text command normalization system (Python, Transformers) | [GitHub](#)

Details

Nizhny Novgorod, Russian Federation

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Links

[GitHub](#)

[Telegram](#)

Technical Skills

C/C++, Python, OpenMP, ONNX Runtime, OpenCV, Shell, Machine Learning, Computer Vision, Profiling, Benchmarking, CI/CD

Languages

Russian — Native

English — B2

Spanish — A1

- Smart voice assistant for drivers (Flet, Python, Pandas) | [GitHub](#)
- Intersection area calculator for convex polygons (Qt, C++) | [GitHub](#)
- Mobile messenger application (Android, Firebase, Java) | [GitHub](#)

Courses and Achievements

Alice & Yandex Smart Devices (Yandex × HSE × MIPT) — Moscow, Russia

April 2026

72-hour intensive on smart-device architecture, embedded systems, hardware/software integration, and deploying ML under resource constraints, completed a team R&D project and successfully defended it | [Link](#)

Hack Lab (Sber × School 21) — Nizhny Novgorod, Russia

November 2025

High-performance computing hackathon focused on parallelization, hardware-level optimization, OpenMP, random number generation, and data integrity mechanisms. Won 1st place as part of the team | [Link](#)

YADRO Winter School "Programming for RISC-V" — Saint Petersburg, Russia

February 2025

Completed an intensive winter school in RISC-V programming covering low-level systems, computer architecture, and practical project work | [Link](#)

All-Russian AI Hackathon "Digital Breakthrough" — Nizhny Novgorod, Russia

September 2023

Developed a GUI application for an intelligent driver assistant in Python using Flet. Built a cross-platform prototype with a clean interface and fast deployment workflow | [Link](#)

Samsung IT-School, Mobile Development — Nizhny Novgorod, Russia

September 2019 — June 2020

Completed coursework in Android development, Java, OOP, algorithms, and data structures. Built a messenger application with Firebase Cloud Storage as a graduation project. Won 1st place in the regional round | [Link](#)

Education

HSE University — Nizhny Novgorod, Russia

September 2026 — Present

Master's Degree in Applied Mathematics and Information Science | Program: Data Mining

HSE University — Nizhny Novgorod, Russia

September 2022 — June 2026

Bachelor's Degree in Applied Mathematics and Information Science | [Diploma](#)

Technical Lyceum #38 — Nizhny Novgorod, Russia

September 2019 — June 2022